

Main voltage regulator

FLASH_SS 2 1 MBR120VLSFT1G R25 10k BTN_SEL

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The diagram illustrates the internal circuitry of the Raspberry Pi RP2350B QFN-80 chip. It shows the power regulation section with VREG_AVDD, VREG_VIN, VREG_LX, and VREG_PGND pins, along with decoupling capacitors C9, C14, C15, C17, C18, C19, C20, C21, C22, C23, C27, C28, C29, and C30. The chip is labeled 'Raspberry Pi RP2350B QFN-80'. The pin numbers 1 through 80 are listed on the left and right sides of the chip. The pins are color-coded: red for power and ground, green for I/O, and blue for SPI and I2S.

The diagram shows the W25Q128BV5 memory chip connected to a QSPI interface. The chip's pins are connected as follows:

- Pin 1 (VCC) to +3V3
- Pin 2 (D(I/O)) to QSPI_S00
- Pin 3 (D(0)) to QSPI_S01
- Pin 4 (D(1)) to QSPI_S02
- Pin 5 (D(2)) to QSPI_S03
- Pin 6 (CLK) to QSPI_SCLK
- Pin 7 (GND) to GND

A 100nF capacitor (C31) is connected between VCC and GND.

The diagram illustrates the electrical connection between a 10-pin connector (J2) and a Micro-SD Card Det1 module. The pins are labeled as follows:

- Pin 1: SD_VDD
- Pin 2: SWCLK
- Pin 3: SD_CS
- Pin 4: SPI_MISO
- Pin 5: SD_VDD
- Pin 6: SPI_SCK
- Pin 7: SPI_MISO
- Pin 8: SWD
- Pin 9: SD_DET
- Pin 10: GND

The Micro-SD Card Det1 module has the following pins:

- DAT2
- DAT3/CD
- CMD
- VDD
- CLK
- VSS
- DAT0
- DAT1
- DET

Resistor values are indicated for the connections:

- R13 10k
- R14 10k
- R15 10k

NB: Pulling SD_DET_MCU hi powers down the SD card.
 Making SD_DET_MCU hi-z powers up the SD card.
 (This way you can reset an already inserted card)

Pinout diagram for the LCD module:

- Power:**
 - +3V3:** Connected to pins 1, 9, 13, 25, 37, and 41.
 - GND:** Connected to pins 2, 10, 14, 26, 38, and 42.
- LCD Pins:**
 - LCD_R5: 8
 - LCD_R4: 9
 - LCD_R3: 10
 - LCD_R2: 11
 - LCD_R1: 12
 - LCD_G5: 14
 - LCD_G4: 15
 - LCD_G3: 16
 - LCD_G2: 17
 - LCD_G1: 18
 - LCD_G0: 19
 - LCD_B5: 20
 - LCD_B4: 21
 - LCD_B3: 22
 - LCD_B2: 23
 - LCD_B1: 24
 - LCD_H5: 26
 - LCD_V5: 27
 - LCD_DE: 28
 - LCD_PCLK: 29
 - LCD_CS: 30
 - SPI_SCK: 31
 - LCD_SPI_MOSI: 32
 - LCD_RST: 33
 - LCD_K: 37
 - LCD_K: 38
 - LCD_K: 39
 - LCD_A: 40

Additional labels: J1, F305-1A7H1-11040-E100, R12 330 LCD_SPI_MOSI_MCU.

The diagram shows the following connections:

- Power Connections:**
 - TP11:** Connected to GND.
 - TP12:** Connected to +3V3.
 - TP7:** Connected to +5V.
- Signal Connections:**
 - LED_A:** Connected to LED_A TP10.
 - ANA_Y:** Connected to ANA_Y TP13.
 - LCD_SPI_MOSI:** Connected to LCD_SPI_MOSI TP15.
 - LED_K:** Connected to LED_K TP15.
 - ANA_X:** Connected to ANA_X TP14.
 - SPI_SCK:** Connected to SPI_SCK TP17.

Legend:

- H1 MountingHole
- H2 MountingHole

H1
MountingHole

H2
MountingHole